

1 Complex Analysis & Contour Integration

1.1 Complex Differentiability, the Cauchy-Riemann Equations, and Analytic Structure

We now extend the calculus of real variables to the complex plane, C . Identifying C topologically with R^2 under the standard Euclidean metric, we analyze functions of a complex variable $z = x + iy$. The requirement of complex differentiability is vastly more rigid than its real counterpart, imposing profound structural symmetry on the component functions.

Let $\Omega \subset C$ be an open set, and let $f : \Omega \rightarrow C$ be a complex-valued function. The function f is **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

where $h \in C$ approaches 0 along any arbitrary path in the complex plane. If f is complex differentiable at every point in an open set Ω , it is said to be **holomorphic** (or analytic) on Ω .

To understand the geometric severity of this definition, we decompose f into its real and imaginary parts: $f(z) = f(x + iy) = u(x, y) + iv(x, y)$, where u and v are real-valued functions of two real variables.

[The Cauchy-Riemann Equations - Necessity] If $f(z) = u(x, y) + iv(x, y)$ is complex differentiable at $z_0 = x_0 + iy_0$, then the first-order partial derivatives of u and v exist at (x_0, y_0) and satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Assume $f'(z_0)$ exists. We evaluate the limit along two orthogonal trajectories.

First, let h approach 0 along the real axis. We write $h = \Delta x + i0$, where $\Delta x \in R$ and $\Delta x \rightarrow 0$. The difference quotient becomes:

$$f'(z_0) = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) + iv(x_0 + \Delta x, y_0) - (u(x_0, y_0) + iv(x_0, y_0))}{\Delta x}$$

Separating the real and imaginary components, we obtain the standard real partial derivatives with respect to x :

$$f'(z_0) = \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

Second, let h approach 0 along the imaginary axis. We write $h = 0 + i\Delta y$, where $\Delta y \in R$ and $\Delta y \rightarrow 0$. The difference quotient evaluates to:

$$f'(z_0) = \lim_{\Delta y \rightarrow 0} \frac{u(x_0, y_0 + \Delta y) + iv(x_0, y_0 + \Delta y) - (u(x_0, y_0) + iv(x_0, y_0))}{i\Delta y}$$

Because $\frac{1}{i} = -i$, we factor the imaginary unit to the numerator and separate the components:

$$f'(z_0) = \lim_{\Delta y \rightarrow 0} \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{\Delta y} - i \lim_{\Delta y \rightarrow 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{\Delta y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Because f is complex differentiable at z_0 , the limit must be strictly path-independent. Equating the real and imaginary parts of the two orthogonal limit evaluations yields:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

This proves the necessity of the Cauchy-Riemann equations.

[Sufficient Conditions for Holomorphy] Let $f(z) = u(x, y) + iv(x, y)$ be defined on an open set $\Omega \subset C$. If the first-order partial derivatives of u and v exist, are continuous on Ω , and satisfy the Cauchy-Riemann equations, then f is holomorphic on Ω .

The proof of sufficiency relies heavily on the total differentiability of mappings in R^2 , leveraging the continuity of the partial derivatives to bound the linear approximation error via the standard Euclidean metric. A profoundly non-trivial consequence of these equations is that any holomorphic function is infinitely differentiable, a fact we will establish via contour integration.